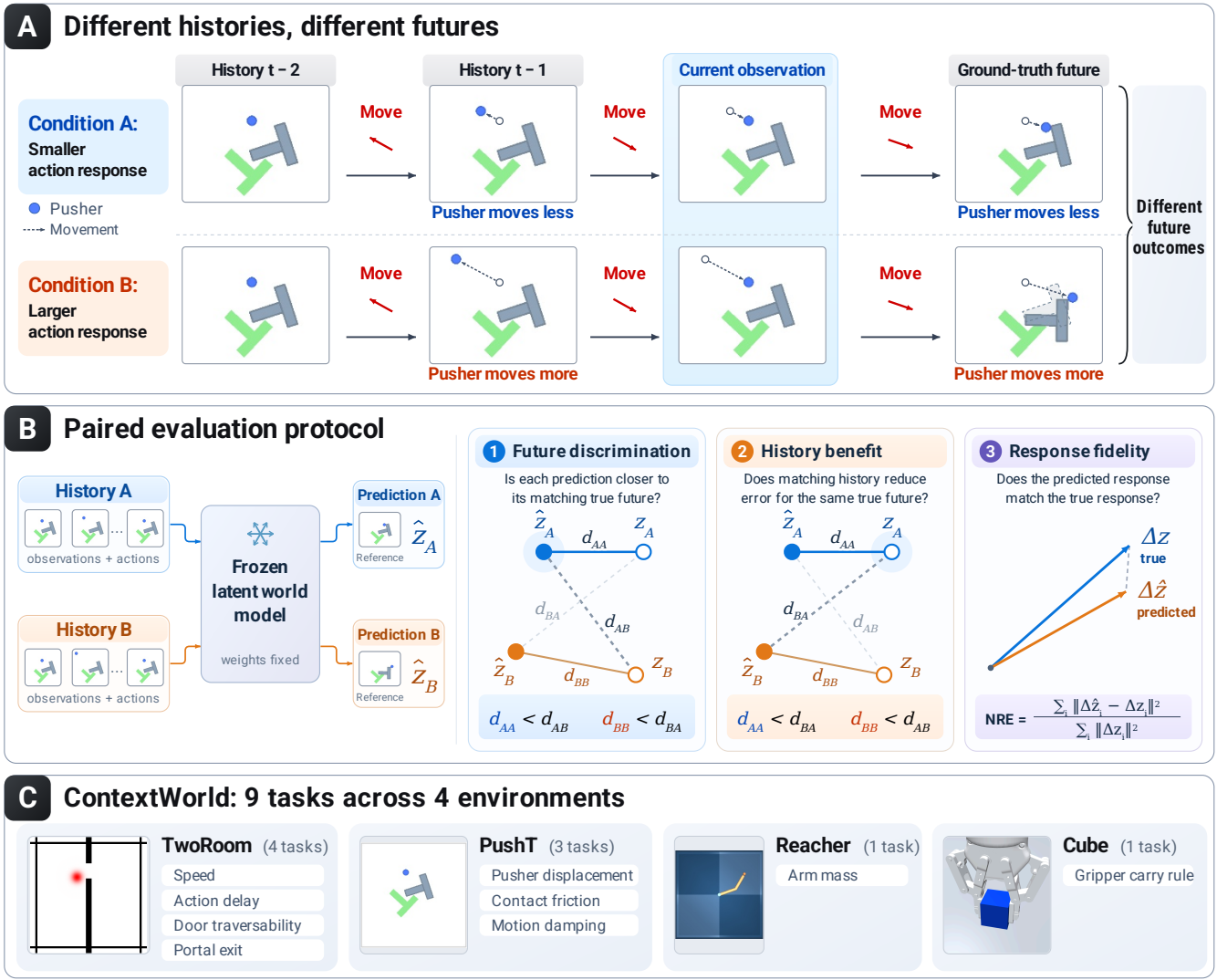

A
Different histories, different futures
History t − 2
History t − 1
Current observation
Ground-truth future
Condition A:
Smaller
action response
Pusher
Movement
Move
Pusher moves less
Pusher moves less
Different
future
outcomes
Condition B:
Larger
action response
Pusher moves more
Pusher moves more
B
Paired evaluation protocol
History A
observations + actions
History B
observations + actions
Frozen
latent world
model
weights fixed
Prediction A
Reference
Prediction B
Reference
1 Future discrimination
Is each prediction closer to
its matching true future?
$d_{AA} < d_{AB}$ $d_{BB} < d_{BA}$
2 History benefit
Does matching history reduce
error for the same true future?
$d_{AA} < d_{BA}$ $d_{BB} < d_{AB}$
3 Response fidelity
Does the predicted response
match the true response?
Δz
true
Δẑ
predicted
NRE =
C
ContextWorld: 9 tasks across 4 environments
TwoRoom (4 tasks)
Speed
Action delay
Door traversability
Portal exit
PushT (3 tasks)
Pusher displacement
Contact friction
Motion damping
Reacher (1 task)
Arm mass
Cube (1 task)
Gripper carry rule